Mathematics: Sets

Course Information

Course Name: Mathematics – Sets

Level: Beginner / Intermediate

Prerequisites: Basic arithmetic and elementary mathematics

Estimated Duration: 6–8 Hours

Course Description

Set theory is a fundamental topic in mathematics that provides a way to collect and organize objects into well-defined groups. In this course, students will learn the basic concepts of sets, methods of representing sets, types of sets, subsets, universal sets, set operations, Venn diagrams, Cartesian products, and important laws of set theory. Students will also learn how to solve problems involving unions, intersections, differences, complements, and cardinality.

Learning Outcomes

After completing this course, students will be able to:

- Define a set and identify its elements.
- Represent sets using different methods.
- Identify different types of sets.
- Determine whether an element belongs to a set.
- Find subsets and proper subsets.
- Understand universal and power sets.
- Perform union, intersection, and difference operations.
- Find the complement of a set.
- Represent sets using Venn diagrams.
- Solve problems involving cardinality.
- Understand Cartesian products.
- Apply important laws of set theory.
- Solve practical problems involving sets.

Lesson 1: Introduction to Sets

Learning Objectives

After this lesson, students will be able to:

- Define a set.
- Identify elements of a set.
- Use set notation.
- Determine whether an object belongs to a set.

What is a Set?

A **set** is a well-defined collection of distinct objects.

The objects in a set are called **elements** or **members** of the set.

For example:

```
A = {1, 2, 3, 4, 5}
```

Here:

- `A` is the set.
- `1, 2, 3, 4, 5` are the elements of the set.

Well-Defined Collection

A collection is well-defined if we can clearly determine whether an object belongs to it.

For example:

```
A = {2, 4, 6, 8}
```

This is a well-defined set because we can clearly determine whether a number belongs to the set.

However:

```
B = {beautiful flowers}
```

may not be well-defined because different people may have different opinions about what is beautiful.

Set Notation

Sets are usually represented using capital letters.

Examples:

```
A = {1, 2, 3}

B = {a, b, c}

C = {red, green, blue}
```

Elements are written inside curly brackets `{ }`.

Membership

The symbol `∈` means "belongs to" or "is an element of."

Example:

```
A = {1, 2, 3, 4}
```

Then:

```
2 ∈ A
```

because 2 belongs to A.

The symbol `∉` means "does not belong to."

```
5 ∉ A
```

because 5 is not an element of A.

Lesson 2: Ways of Representing Sets

There are several ways to represent a set.

1. Roster or Tabular Form

In roster form, all elements are listed inside curly brackets.

Example:

```
A = {2, 4, 6, 8, 10}
```

---

2. Set-Builder Form

In set-builder form, we describe the rule that determines the elements.

Example:

```
A = {x | x is an even number less than 12}
```

The vertical bar `|` can be read as "such that."

---

3. Descriptive Form

A set can also be described using words.

Example:

```
A = {the first five positive even integers}
```

Therefore:

```
A = {2, 4, 6, 8, 10}
```

Lesson 3: Types of Sets

Sets can be classified into different types.

Empty Set

An **empty set** contains no elements.

It is represented by:

```
∅
```

or:

```
{ }
```

Example:

The set of natural numbers less than 0:

```
A = ∅
```

Singleton Set

A set containing exactly one element is called a singleton set.

Example:

```
A = {5}
```

Finite Set

A set containing a limited number of elements is called a finite set.

Example:

```
A = {1, 2, 3, 4, 5}
```

Infinite Set

A set containing infinitely many elements is called an infinite set.

Example:

```
N = {1, 2, 3, 4, 5, ...}
```

Equal Sets

Two sets are equal if they contain exactly the same elements.

Example:

```
A = {1, 2, 3}

B = {3, 2, 1}
```

Therefore:

```
A = B
```

The order of elements does not matter in a set.

Equivalent Sets

Two sets are equivalent if they contain the same number of elements.

Example:

```
A = {1, 2, 3}

B = {a, b, c}
```

Both sets contain three elements, so they are equivalent.

Lesson 4: Subsets

Definition

A set `A` is a subset of set `B` if every element of `A` is also an element of `B`.

The symbol is:

```
A ⊆ B
```

Example:

```
A = {1, 2}

B = {1, 2, 3, 4}
```

Therefore:

```
A ⊆ B
```

Proper Subset

If `A` is a subset of `B` but `A` is not equal to `B`, then `A` is a proper subset of `B`.

The symbol is:

```
A ⊂ B
```

Example:

```
A = {1, 2}

B = {1, 2, 3}
```

Therefore:

```
A ⊂ B
```

Number of Subsets

If a set contains `n` elements, the number of subsets is:

```
2^n
```

Example:

If:

```
A = {1, 2, 3}
```

Then `n = 3`.

Number of subsets:

```
2^3 = 8
```

Lesson 5: Power Set

The **power set** of a set is the set containing all possible subsets of that set.

It is represented by:

```
P(A)
```

Example:

```
A = {1, 2}
```

The subsets are:

```
∅
{1}
{2}
{1, 2}
```

Therefore:

```
P(A) = {∅, {1}, {2}, {1, 2}}
```

Since A has two elements:

```
|P(A)| = 2^2 = 4
```

Lesson 6: Universal Set

The **universal set** contains all objects under consideration in a particular problem.

It is usually represented by:

```
U
```

Example:

Suppose we are discussing numbers from 1 to 10.

Then:

```
U = {1,2,3,4,5,6,7,8,9,10}
```

If:

```
A = {2,4,6,8,10}
```

then A is a subset of U.

Lesson 7: Set Operations

Set operations allow us to combine or compare sets.

The main operations are:

- Union
- Intersection
- Difference
- Complement

Lesson 8: Union of Sets

The **union** of two sets contains all elements that belong to either set.

The symbol is:

```
A ∪ B
```

Example:

```
A = {1, 2, 3}

B = {3, 4, 5}
```

Therefore:

```
A ∪ B = {1, 2, 3, 4, 5}
```

The repeated element `3` is written only once.

Lesson 9: Intersection of Sets

The **intersection** of two sets contains elements common to both sets.

The symbol is:

```
A ∩ B
```

Example:

```
A = {1, 2, 3}

B = {3, 4, 5}
```

Therefore:

```
A ∩ B = {3}
```

Disjoint Sets

Two sets are disjoint if they have no elements in common.

Example:

```
A = {1, 2, 3}

B = {4, 5, 6}
```

Therefore:

```
A ∩ B = ∅
```

Lesson 10: Difference of Sets

The difference between two sets contains elements that belong to the first set but not the second set.

The symbol is:

```
A - B
```

Example:

```
A = {1, 2, 3, 4}

B = {3, 4, 5, 6}
```

Therefore:

```
A - B = {1, 2}
```

Similarly:

```
B - A = {5, 6}
```

Lesson 11: Complement of a Set

The complement of a set contains all elements in the universal set that are not in the given set.

The complement of A can be written as:

```
A'
```

or:

```
Aᶜ
```

Example:

```
U = {1,2,3,4,5,6}

A = {2,4,6}
```

Therefore:

```
Aᶜ = {1,3,5}
```

Lesson 12: Venn Diagrams

A **Venn diagram** is a visual representation of sets.

Sets are usually represented using circles inside a rectangle representing the universal set.

For example:

```
          Universal Set U

      ┌──────────────────────────┐
      |                          |
      |       A       B          |
      |     (   )   (   )        |
      |        \___/             |
      |                          |
      └──────────────────────────┘
```

Venn diagrams can be used to visualize:

- Union
- Intersection
- Difference
- Complement
- Disjoint sets

Lesson 13: Union and Intersection Using Venn Diagrams

Suppose:

```
A = {1,2,3}

B = {3,4,5}
```

The common element is:

```
3
```

Therefore:

```
A ∩ B = {3}
```

The complete collection is:

```
A ∪ B = {1,2,3,4,5}
```

In a Venn diagram:

- The overlapping region represents the intersection.
- Both circles together represent the union.

Lesson 14: Cardinality of Sets

The **cardinality** of a set is the number of elements in the set.

It is represented by:

```
|A|
```

Example:

```
A = {2,4,6,8}
```

Therefore:

```
|A| = 4
```

Cardinality of Union

For two finite sets:

```
|A ∪ B| = |A| + |B| - |A ∩ B|
```

Example:

If:

```
|A| = 20
|B| = 15
|A ∩ B| = 5
```

Then:

```
|A ∪ B| = 20 + 15 - 5
```

Therefore:

```
|A ∪ B| = 30
```

Lesson 15: Cartesian Product

The **Cartesian product** of two sets is the set of all ordered pairs where the first element comes from the first set and the second element comes from the second set.

The Cartesian product is written as:

```
A × B
```

Example:

```
A = {1,2}

B = {a,b}
```

Then:

```
A × B =
{(1,a), (1,b), (2,a), (2,b)}
```

Number of Ordered Pairs

If:

```
|A| = m
```

and:

```
|B| = n
```

then:

```
|A × B| = m × n
```

Example:

```
|A| = 2
|B| = 3
```

Therefore:

```
|A × B| = 2 × 3 = 6
```

Lesson 16: Laws of Set Theory

Set theory has several important laws.

Commutative Laws

For union:

```
A ∪ B = B ∪ A
```

For intersection:

```
A ∩ B = B ∩ A
```

Associative Laws

For union:

```
(A ∪ B) ∪ C = A ∪ (B ∪ C)
```

For intersection:

```
(A ∩ B) ∩ C = A ∩ (B ∩ C)
```

Distributive Laws

```
A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)
```

```
A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)
```

Identity Laws

```
A ∪ ∅ = A
```

```
A ∩ U = A
```

Domination Laws

```
A ∪ U = U
```

```
A ∩ ∅ = ∅
```

Idempotent Laws

```
A ∪ A = A
```

```
A ∩ A = A
```

Lesson 17: De Morgan's Laws

De Morgan's laws are important rules involving complements.

First Law

```
(A ∪ B)ᶜ = Aᶜ ∩ Bᶜ
```

The complement of a union is equal to the intersection of the complements.

Second Law

```
(A ∩ B)ᶜ = Aᶜ ∪ Bᶜ
```

The complement of an intersection is equal to the union of the complements.

Lesson 18: Applications of Sets

Sets are used in many real-world and mathematical applications.

Examples include:

- Student enrollment
- Database systems
- Search engines
- Probability
- Statistics
- Computer science
- Classification
- Logic
- Data analysis

Example: Student Courses

Suppose:

```
A = Students studying Mathematics
```

and:

```
B = Students studying Computer Science
```

Then:

```
A ∩ B
```

represents students studying both Mathematics and Computer Science.

Similarly:

```
A ∪ B
```

represents students studying Mathematics, Computer Science, or both.

Lesson 19: Solved Problems

Problem 1

Given:

```
A = {1,2,3,4}
B = {3,4,5,6}
```

Find:

```
A ∪ B
```

Solution

Combine all unique elements:

```
A ∪ B = {1,2,3,4,5,6}
```

Problem 2

Given:

```
A = {1,2,3,4}
B = {3,4,5,6}
```

Find:

```
A ∩ B
```

Solution

The common elements are `3` and `4`.

Therefore:

```
A ∩ B = {3,4}
```

Problem 3

Given:

```
A = {1,2,3,4,5}
B = {2,4}
```

Determine whether B is a subset of A.

Solution

Every element of B appears in A.

Therefore:

```
B ⊆ A
```

Problem 4

Given:

```
U = {1,2,3,4,5,6,7,8}
A = {2,4,6,8}
```

Find:

```
Aᶜ
```

Solution

Elements in U that are not in A are:

```
Aᶜ = {1,3,5,7}
```

Problem 5

If:

```
|A| = 25
|B| = 18
|A ∩ B| = 7
```

Find:

```
|A ∪ B|
```

Solution

Use:

```
|A ∪ B| = |A| + |B| - |A ∩ B|
```

Substitute the values:

```
|A ∪ B| = 25 + 18 - 7
```

Therefore:

```
|A ∪ B| = 36
```

Practice Exercises

Exercise 1

Given:

```
A = {2,4,6,8}
B = {4,8,12,16}
```

Find:

1. `A ∪ B`
2. `A ∩ B`
3. `A - B`
4. `B - A`

Exercise 2

Given:

```
A = {a,b,c}
```

Find all subsets of A.

Exercise 3

If a set contains 5 elements, determine the number of subsets.

Use:

```
2^n
```

Exercise 4

Given:

```
U = {1,2,3,4,5,6,7,8,9,10}

A = {2,4,6,8,10}
```

Find:

```
Aᶜ
```

Exercise 5

Given:

```
A = {1,2}
B = {x,y,z}
```

Find:

```
A × B
```

Mini Project

Student Course Enrollment Analysis

A university wants to analyze students enrolled in different courses.

Suppose:

```
A = Students enrolled in Mathematics

B = Students enrolled in Computer Science
```

Students should be represented using unique identifiers.

The project should determine:

1. Students studying Mathematics.
2. Students studying Computer Science.
3. Students studying both subjects.
4. Students studying either subject.
5. Students studying Mathematics but not Computer Science.
6. Total number of students.
7. Results using a Venn diagram.

Concepts to Use

The project should demonstrate:

- Sets
- Subsets
- Union

- Intersection
- Difference
- Complement
- Cardinality
- Venn diagrams

Final Quiz

1. What is a set?
2. What is an element of a set?
3. What does the symbol `∈` mean?
4. What does the symbol `∉` mean?
5. What is an empty set?
6. What is a singleton set?
7. What is the difference between finite and infinite sets?
8. What are equal sets?
9. What are equivalent sets?
10. What is a subset?
11. What is a proper subset?
12. How many subsets does a set with `n` elements have?
13. What is a power set?
14. What is a universal set?
15. What is the union of two sets?
16. What is the intersection of two sets?
17. What is the difference between two sets?
18. What is the complement of a set?
19. What is a Venn diagram?
20. What is the cardinality of a set?
21. What is the formula for the cardinality of a union?
22. What is a Cartesian product?
23. What are De Morgan's laws?
24. What are disjoint sets?
25. Give one real-world application of set theory.

Course Summary

After completing this course, students should understand the fundamental concepts of set theory and be able to solve basic and intermediate problems involving sets.

The major concepts covered are:

- Definition of sets
- Elements and membership
- Set notation
- Methods of representing sets
- Types of sets
- Subsets

- Proper subsets
- Power sets
- Universal sets
- Union
- Intersection
- Difference
- Complement
- Venn diagrams
- Cardinality
- Cartesian products
- Laws of set theory
- De Morgan's laws
- Applications of sets
- Solving set problems

Next Step

After completing Sets, students can continue with related mathematical topics such as **Relations and Functions, Logic, Probability, or Basic Counting Principles**.